Model Card: AI-CRS — Preoperative Sinus CT Models for Predicting Surgical Outcomes in Chronic Rhinosinusitis

Version 1.0, 2026-11-07

Model Metadata

Field	Value
language	English
license	MIT (to be confirmed against the LICENSE file added to the repository)
library_name	PyTorch
pipeline_tag	Image-classification
tags	Medical; radiology; otolaryngology; chronic-rhinosinusitis; computed-tomography; radiomics; deep-learning; prognostic-model; binary-classification
metrics	AUC; Caliration; DCA; accuracy; sensitivity; specificity; Dice; IoU

1. Model Details

Item	Description
Model name	AI-CRS model suite
Version / date	V1.0, 2026-11-07
Model type	Two-stage pipeline: (i) 3D CT segmentation; (ii) binary prognostic classification
Segmentation model	nnUNetV2 (self-configuring); four configurations trained — 2D, 3D low-resolution, 3D full-resolution, 3D cascade full-resolution
Prediction models	Clinical model (logistic regression, blood eosinophil count); CT-score model (logistic regression, Lund-Mackay + GOSS score); CT-radiomic model (XGBoost on hand-crafted radiomic features); CT-DL model (3dResNet on automatically learned features); Combined models (CT features + eosinophil count)
Task	Predict uncontrolled CRS after endoscopic sinus surgery (ESS) from a preoperative sinus CT
Input	Non-contrast sinus CT (DICOM), 512 × 512 matrix, 150 × 150 mm field of view, slice thickness ≤ 2 mm, acquired 1-3 days before surgery
Output	Sinus segmentation mask Probability of uncontrolled disease at ≥ 12 months postoperatively; binary label at the Youden's index threshold
Developers	Ke-Zhang Zhu, et al.; corresponding author Zheng Liu

Contact	kzzhuu@foxmail.com; zhengliuent@hotmail.com
Code	https://github.com/Yukeyuzhang/AI_CRS.git
Deployment	https://www.tongjicrsdata.com (cloud platform, one-click inference)
Associated publication	Development and Evaluation of Radiomic and Deep Learning Models Based on Preoperative Sinus CT Images for Predicting Surgical Outcomes in Chronic Rhinosinusitis: A Nationwide Multicenter Study
Clinical trial registration	ClinicalTrials.gov NCT05983003
Ethics approval	Research Ethics Committee, Wuhan Tongji Hospital: TJ-IRB20220666 Conducted per the Declaration of Helsinki

Outcome definition

Per EPOS 2020, disease was classified as **uncontrolled** when ≥ 3 of the following 7 elements were present **in the last month** of follow-up: (i) nasal blockage; (ii) rhinorrhea or postnasal drip; (iii) facial pain or pressure; (iv) smell dysfunction; (v) sleep disturbance or fatigue; (vi) endoscopic finding of diseased mucosa (if available); (vii) symptoms persisting despite rescue treatment (oral broad-spectrum antibiotics and/or corticosteroids) in the past 6 months. Symptom presence was defined as a visual analogue scale (VAS) score > 5 on most days per week.

2. Intended Use

Intended users and setting

Otolaryngologists and rhinology teams in secondary/tertiary hospitals, for **preoperative risk stratification** of adults with chronic rhinosinusitis (with or without nasal polyps) who have persistent symptoms despite medical therapy and are scheduled for ESS.

Intended decisions supported

- Counselling patients on the probability of persistent/uncontrolled disease after surgery.
- Population-level triage where automated, high-throughput assessment is required.

Out-of-scope uses

- **Not a diagnostic tool** for CRS, and not a substitute for EPOS 2020 diagnostic criteria, nasal endoscopy, or clinical judgement.
- Not validated for the conditions excluded from the study: fungal sinusitis, odontogenic maxillary sinusitis, choanal polyps, cystic fibrosis, vasculitis, primary ciliary dyskinesia, immunodeficiency, prior cranial trauma, head and neck tumours.
- Not validated for CT slice thickness > 2 mm, or for scans with metal/motion artefacts or incomplete sinus coverage.

- Not validated for patients managed non-surgically, for acute asthma exacerbation or acute upper-respiratory infection within 1 month, or within the corticosteroid washout windows used for enrolment (oral ≥ 3 months; intranasal ≥ 1 month).
- Not a treatment-recommendation engine; outputs are advisory and require clinician review.

3. Factors, Limitations, Biases and Ethical Considerations

Relevant factors

Factor	Status in this study
Sex	Male predominance: 1,880/2,755 (68.2%) overall; 863/1,242 (69.5%) train-val, 339/481 (70.5%) inter-test, 678/1,032 (65.7%) exter-test. Sex was recorded from medical records, was not an enrolment criterion, and was not used for stratified subgroup analysis.
Gender identity	Not collected.
Age	Recorded and reported (Table 1); no age-stratified performance analysis was performed.
Scanner / protocol	Kernel, kVp and mAs varied by centre; stratified analyses showed stable performance across these parameters.
Surgical extent	Stable performance in both limited and extensive ESS subgroups.
Outcome prevalence	Uncontrolled rates differ between development (36.63%) and external test (43.12%)

Known limitations

(1) **Retrospective design.** All data were collected retrospectively (January 2015-August 2023). Prospective validation is required before clinical deployment.

(2) **Population narrowness.** Exclusively Chinese surgical CRS patients; no race/ethnicity data. Transportability to other health systems is unestablished.

(3) **Segmentation dependency of the radiomic model.** Radiomic features are extracted from threshold-based semi-automatic ROIs delineated by local radiologists (3D Slicer 4.12.0). This is labour-intensive and operator-dependent; performance under fully automatic segmentation is not guaranteed for this model.

(4) **Opacity of the DL model.** The 3dResNet model is a black box.

(5) **Biological interpretation is exploratory.** Radiomic–proteomic/histologic correlations were derived from a 128-patient subset and have not been externally validated.

(6) **No interventional evidence.** Randomized controlled trials are needed to establish that model-guided management improves patient outcomes.

Ethical considerations

- Data were anonymized at source; images were de-identified before segmentation and scoring, removing clinical identifiers from readers.

- Data are **not openly downloadable**: de-identified tabular data are released for non-commercial academic use under a formal data usage agreement, on request to the corresponding author (response within approximately 2 months). A 50-patient anonymized demo dataset is available for reproducibility.
- **Risk of automation bias**: a false-negative prediction could delay intensified postoperative treatment. Model output must not override clinical assessment.

4. Training Data

Cohort	Source	n	Uncontrolled, n (%)	Role
Train-val	Wuhan Tongji Hospital	1,242	455 (36.63%)	Model development, feature selection, hyperparameter tuning
Internal-test (inter-test)	Wuhan Tongji Hospital	481	171 (35.55%)	Held-out internal evaluation
External-test (exter-test)	11 other centres, pooled into 8 cohorts A-H	1,032	445 (43.12%)	Held-out external evaluation
Total enrolled	12 centres, China	2,755	1,071 (38.87%)	—

- **Screening**: 6,792 ESS patients screened (Jan 2015 – Aug 2023); 2,755 enrolled after clinical and radiological exclusions.
- **Split**: random **7:3** division of the Tongji cohort into train-val and inter-test. External cohorts were held out **by hospital**; centres contributing < 124 patients (1/10 of train-val size) were pooled (Cohort D, n = 130; Cohort H, n = 130).
- **Leakage control**: all model development and parameter tuning were performed exclusively on the train-val cohort, with no access to any test data. Test cohorts were used only for final evaluation.
- **Follow-up**: minimum 12 months; median 38 months (IQR 28–59).
- **External cohorts**:

  A — Wuhan Central Hospital (126);

  B — Shiyan Taihe Hospital (130);

  C — Xiangyang Central Hospital (135);

  D — Hubei Yangxin + Hubei Xinhua + Wuhan Tongren (130);

  E — Shanxi Bethune Hospital (124);

  F — Zhengzhou First Hospital (133);

  G — Nanchang First Hospital (124);

  H — Clifford + Longgang Central Hospital (130).
- **Biological subset**: tissue samples from 128 Tongji patients (90 train-val, 38 inter-test) for LC-MS/MS proteomics, Bio-Plex cytokine assay and H&E histology.

5. Training Parameters and Experimental Information

Preprocessing

- Resampling to **1 × 1 × 1 mm** isotropic voxels (sitkBSpline interpolation); **Z-score** intensity normalization; bone window (WW 2000 HU / WL 200 HU) and soft-tissue window (WW 350 HU) for scoring.
- ROI = entire nasal cavity and paranasal sinuses; sub-ROIs: whole ROI, air-cavity ROI, soft-tissue ROI (threshold segmentation).

Segmentation (nnUNetV2)

Parameter	Value
Framework	nnUNetV2 2.5.1
Configurations	2D, 3D low-resolution, 3D full-resolution, 3D cascade full-resolution
Cross-validation	5-fold, predictions ensembled per configuration
Epochs	1,000
Voxel spacing / patch size / batch size	Auto-configured by nnUNetV2 after axis reorientation
Supervision	Semi-automatic masks as reference labels
Model selection	Best-performing single configuration or ensemble, chosen on train-val
Deep-learning stack	PyTorch 2.4.1, CUDA 12.6
Hardware	NVIDIA GeForce RTX 4090

Radiomic feature extraction and selection

Step	Setting
Package	PyRadiomics 3.1.0, IBSI-compliant
Discretization	Fixed bin width 15 HU
Feature classes	First-order (soft-tissue ROI); shape (whole + air-cavity ROI); texture GLCM / GLRLM / GLSZM / NGTDM / GLDM (soft-tissue ROI)
Filters	Wavelet (Coiflet-1), LoG (σ 0.5-2.5 mm), 3D LBP, gradient
Additional indices	Soft-tissue/total-sinus volume ratio; air/total-sinus volume ratio
Extracted	1,981 features
Reproducibility filter	Intra- and inter-observer ICC > 0.80 → 1,802 retained (inter-observer: 30 cases, 2 radiologists; intra-observer: 50 cases re-segmented after 3 months)

Prediction models

Parameter	CT-radiomic (XGBoost)	CT-DL (3dResNet)
Candidates compared	XGBoost, LASSO, RF, SVM, LR (backward stepwise)	3dResNet, 3dDenseNet, 3dVGG, 3dAlexNet, 3dViT, 3dSwinT
Selected	XGBoost (highest train-val AUC)	3dResNet (highest train-val AUC)

Software environment

Python 3.10.15; R 4.4.2; PyTorch 2.4.1; CUDA 12.6; nnUNetV2 2.5.1; PyRadiomics 3.1.0; 3D Slicer 4.12.0; Adobe Illustrator 2025. Full package/function/version list: **Supplementary Table 12** of the manuscript.

6. Evaluation Results

Metrics and rationale

- **Discrimination**: ROC AUC with 95% CI; between-model comparison by **DeLong's test**.
- **Calibration**: calibration curves + **Hosmer–Lemeshow test**.
- **Classification**: accuracy, sensitivity, specificity (threshold 0.5), confusion matrices.
- **Clinical utility**: decision curve analysis (DCA), net reclassification improvement (NRI), integrated discrimination improvement (IDI).

Segmentation agreement (automatic vs. semi-automatic reference)

Cohort	Dice, median (IQR)	IoU, median (IQR)
Train-val (n = 1,242)	0.955 (0.943-0.961)	0.914 (0.892-0.924)
Inter-test (n = 481)	0.955 (0.942-0.961)	0.914 (0.890-0.925)
Entire exter-test (n = 1,032)	0.950 (0.931-0.959)	0.905 (0.871-0.921)
Individual exter-test cohorts A–H	0.947-0.953	0.899-0.909

Prognostic discrimination (primary results, manuscript Table 2)

Cohort	Model	AUC (95% CI)	Accuracy	Sensitivity	Specificity
Train-val (n = 1,242)	Clinical	0.665 (0.634-0.697)	0.628	0.644	0.619
Train-val (n = 1,242)	CT-score	0.729 (0.700-0.758)	0.680	0.626	0.712
Train-val (n = 1,242)	CT-radiomic (XGBoost)	0.875 (0.856-0.895)	0.816	0.734	0.864
Train-val (n = 1,242)	CT-DL (3dResNet)	0.851 (0.829-0.873)	0.783	0.756	0.799
Inter-test (n = 481)	Clinical	0.664 (0.614-0.714)	0.655	0.708	0.626
Inter-test (n = 481)	CT-score	0.717 (0.668-0.766)	0.676	0.661	0.684
Inter-test (n = 481)	CT-radiomic	0.836 (0.799-0.873)	0.775	0.708	0.813

Inter-test (n = 481)	CT-DL	0.807 (0.766-0.849)	0.753	0.731	0.765
Entire exter-test (n = 1,032)	Clinical	0.650 (0.617-0.684)	0.606	0.647	0.574
Entire exter-test (n = 1,032)	CT-score	0.718 (0.687-0.750)	0.671	0.652	0.685
Entire exter-test (n = 1,032)	CT-radiomic	0.863 (0.840-0.885)	0.779	0.800	0.763
Entire exter-test (n = 1,032)	CT-DL	0.827 (0.801-0.852)	0.734	0.735	0.733

Combined models (CT features + blood eosinophil count)

Model	**AUC range across train-val / inter-test / exter-test**
Combined 1 (CT-radiomic + eosinophil)	0.853-0.881
Combined 2 (CT-DL + eosinophil)	0.832-0.858
Combined 3 (CT-score + eosinophil)	0.732-0.753

Calibration, clinical utility and robustness

- **Calibration**: Hosmer–Lemeshow **P > 0.05** in all cohorts (train-val, inter-test, entire and individual exter-test); calibration curves close to the 45° ideal line.
- **Clinical utility**: DCA showed higher net benefit than CT-score and clinical models across a wide threshold-probability range in all cohorts; **NRI > 0.35** and **IDI > 0.1** versus both comparators.
- **Subgroup robustness**: stable across limited vs. extensive ESS, and across CT kernel, kVp and mAs.

Interpretability

- **SHAP** on the radiomic model
- **Grad-CAM** on 3dResNet
- **Multi-omics**: unsupervised consensus clustering of radiomic features yielded **4 stable clusters** (k = 4 by elbow, consensus CDF, item-tracking and consensus-matrix criteria), associated with distinct clinical, proteomic (LC-MS/MS, Bio-Plex) and histopathological profiles.

7. Caveats and Recommendations

(1) Use the **CT-radiomic model** when interpretability and individualized risk assessment are priorities and manual segmentation is feasible; use the **CT-DL model** for automated, high-throughput workflows.

(2) Recalibrate or re-select the operating threshold when local outcome prevalence differs materially from 36-43%.

(3) Verify CT acquisition meets the study's quality criteria (slice ≤ 2 mm, complete sinus coverage, no significant artefact) before inference.

(4) Treat outputs as **advisory**. Do not use a single prediction to withhold indicated surgery or postoperative therapy.